PCA and Dimensionality Reduction

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: PCA & Dimensionality Reduction Practice Exercises**

Consider a dataset with n data points $\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n\}$, where each $\mathbf{x}_i \in \mathbb{R}^p$. The sample covariance matrix is given by $\mathbf{S} = \frac{1}{n-1}\sum_{i=1}^{n}(\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^T$, where $\bar{\mathbf{x}} = \frac{1}{n}\sum_{i=1}^{n}\mathbf{x}_i$ is the sample mean. Suppose PCA is performed on this dataset, resulting in principal components defined by the eigenvectors of $\mathbf{S}$ and corresponding variances given by the eigenvalues. Let $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p$ be the orthonormal eigenvectors of $\mathbf{S}$ with associated eigenvalues $\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_p$. The first principal component captures the direction of maximum variance, λ_1, and the second principal component captures the next largest variance, λ_2, orthogonal to the first. If we decide to reduce the dimensionality from p to k by keeping only the first k principal components, the proportion of the total variance retained is given by $\frac{\sum_{i=1}^{k}\lambda_i}{\sum_{i=1}^{p}\lambda_i}$. Suppose for a particular dataset, the eigenvalues of the covariance matrix are $\{10.5, 5.2, 2.1, 0.8, 0.4\}$. If we aim to retain at least 90% of the total variance, what is the minimum number of principal components required? Show your complete algebraic work.

Final Answer: 3. The total variance in the dataset is the sum of all eigenvalues: $\sum_{i=1}^{p}\lambda_i = 10.5 + 5.2 + 2.1 + 0.8 + 0.4 = 19.0$. We need to find the minimum k such that $\frac{\sum_{i=1}^{k}\lambda_i}{\sum_{i=1}^{p}\lambda_i} \geq 0.90$. Let's check for different values of k: For $k = 1$: The variance retained is $\frac{10.5}{19.0} \approx 0.553$, which is less than 0.90. For $k = 2$: The variance retained is $\frac{10.5+5.2}{19.0} = \frac{15.7}{19.0} \approx 0.826$, which is less than 0.90. For.

2 **[Foundational]** • **Target: Data Centering and the Covariance Matrix**

Consider a dataset of observations $\{x_i\}_{i=1}^{n}$ where each x_i is a 2-dimensional vector. The sample covariance matrix for this dataset is given by: $S = (4, -2; -2, 1)$ Principal Component Analysis (PCA) is applied to this dataset. The first principal component corresponds to the direction with the largest variance. Let the direction vector of the first principal component be $\mathbf{v}_1 = (a; b)$. The variance explained by this principal component is given by the corresponding eigenvalue λ_1. If $\mathbf{v}_1$ is a unit vector, what is the value of $a^2 + b^2$? Show your complete algebraic work.

Final Answer: 1. The first principal component direction vector $\mathbf{v}_1$ is an eigenvector of the covariance matrix S. The variance explained by this component is the corresponding eigenvalue λ_1. The problem states that $\mathbf{v}_1 = (a; b)$ is a unit vector. By definition, a unit vector has a magnitude of 1. The squared magnitude of a vector $(a; b)$ is $a^2 + b^2$. Therefore, if $\mathbf{v}_1$ is a unit vector, its squared magnitude must be 1, which means $a^2 + b^2 = 1$.

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3 [Foundational] • Target: Eigenvalues, Eigenvectors and Variance Maximization

A 3D dataset has covariance matrix eigenvalues $\lambda_1 = 62$, $\lambda_2 = 24$, and $\lambda_3 = 14$. What percentage of total variance is explained by the first principal component PC_1? Show your complete algebraic work.

Final Answer: 62.0%. Total variance is $\sum \lambda_i = 62 + 24 + 14 = 100$. The proportion explained by PC_1 is $\frac{\lambda_1}{\sum \lambda_i} = \frac{62}{100} = 62.0\%$.

4 [Foundational] • Target: Explained Variance and Scree Analysis

A 3D dataset has covariance matrix eigenvalues $\lambda_1 = 64$, $\lambda_2 = 23$, and $\lambda_3 = 13$. What percentage of total variance is explained by the first principal component PC_1? Show your complete algebraic work.

Final Answer: 64.0%. Total variance is $\sum \lambda_i = 64 + 23 + 13 = 100$. The proportion explained by PC_1 is $\frac{\lambda_1}{\sum \lambda_i} = \frac{64}{100} = 64.0\%$.

5 [Foundational] • Target: SVD Connection and Applications

A 3D dataset has covariance matrix eigenvalues $\lambda_1 = 66$, $\lambda_2 = 22$, and $\lambda_3 = 12$. What percentage of total variance is explained by the first principal component PC_1? Show your complete algebraic work.

Final Answer: 66.0%. Total variance is $\sum \lambda_i = 66 + 22 + 12 = 100$. The proportion explained by PC_1 is $\frac{\lambda_1}{\sum \lambda_i} = \frac{66}{100} = 66.0\%$.

6 [Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises

A 3D dataset has covariance matrix eigenvalues $\lambda_1 = 68$, $\lambda_2 = 21$, and $\lambda_3 = 11$. What percentage of total variance is explained by the first principal component PC_1? Show your complete algebraic work.

Final Answer: 68.0%. Total variance is $\sum \lambda_i = 68 + 21 + 11 = 100$. The proportion explained by PC_1 is $\frac{\lambda_1}{\sum \lambda_i} = \frac{68}{100} = 68.0\%$.

7 [Foundational] • Target: Data Centering and the Covariance Matrix

A 3D dataset has covariance matrix eigenvalues $\lambda_1 = 70$, $\lambda_2 = 20$, and $\lambda_3 = 10$. What percentage of total variance is explained by the first principal component PC_1? Show your complete algebraic work.

Final Answer: 70.0%. Total variance is $\sum \lambda_i = 70 + 20 + 10 = 100$. The proportion explained by PC_1 is $\frac{\lambda_1}{\sum \lambda_i} = \frac{70}{100} = 70.0\%$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Eigenvalues, Eigenvectors and Variance Maximization

A 3D dataset has covariance matrix eigenvalues $\lambda_1 = 72$, $\lambda_2 = 19$, and $\lambda_3 = 9$. What percentage of total variance is explained by the first principal component PC_1? Show your complete algebraic work.

Final Answer: 72.0%. Total variance is $\sum \lambda_i = 72 + 19 + 9 = 100$. The proportion explained by PC_1 is $\frac{\lambda_1}{\sum \lambda_i} = \frac{72}{100} = 72.0\%$.

9 [Intermediate] • Target: Explained Variance and Scree Analysis

A 3D dataset has covariance matrix eigenvalues $\lambda_1 = 74$, $\lambda_2 = 18$, and $\lambda_3 = 8$. What percentage of total variance is explained by the first principal component PC_1? Show your complete algebraic work.

Final Answer: 74.0%. Total variance is $\sum \lambda_i = 74 + 18 + 8 = 100$. The proportion explained by PC_1 is $\frac{\lambda_1}{\sum \lambda_i} = \frac{74}{100} = 74.0\%$.

10 [Intermediate] • Target: SVD Connection and Applications

A 3D dataset has covariance matrix eigenvalues $\lambda_1 = 76$, $\lambda_2 = 17$, and $\lambda_3 = 7$. What percentage of total variance is explained by the first principal component PC_1? Show your complete algebraic work.

Final Answer: 76.0%. Total variance is $\sum \lambda_i = 76 + 17 + 7 = 100$. The proportion explained by PC_1 is $\frac{\lambda_1}{\sum \lambda_i} = \frac{76}{100} = 76.0\%$.

11 [Intermediate] • Target: PCA & Dimensionality Reduction Practice Exercises

Consider a dataset with two features, X_1 and X_2. The covariance matrix for this dataset is given by: $\Sigma = (10, 5; 5, 20)$ What is the variance of the first feature, X_1? Show your complete algebraic work.

Final Answer: 10. The covariance matrix Σ for a dataset with two features X_1 and X_2 is structured as follows: $\Sigma = (Cov(X_1, X_1), Cov(X_1, X_2); Cov(X_2, X_1), Cov(X_2, X_2))$ By definition, the variance of a random variable is the covariance of the variable with itself. Therefore, $Var(X_1) = Cov(X_1, X_1)$ and $Var(X_2) = Cov(X_2, X_2)$. In the given covariance matrix: $\Sigma = (10, 5; 5, 20)$ The element in the first row and first column represents $Cov(X_1, X_1)$, whi.

12 [Intermediate] • Target: Data Centering and the Covariance Matrix

A centered feature matrix $\mathbf{X}$ has shape $N \times 14$. What is the size and symmetry property of the sample covariance matrix $\mathbf{\Sigma} = \frac{1}{N-1}\mathbf{X}^T\mathbf{X}$? Show your complete algebraic work.

Final Answer: 14×14 symmetric matrix. Since $\mathbf{X}$ has 14 columns, $\mathbf{X}^T$ is $14 \times N$ and $\mathbf{X}^T\mathbf{X}$ is 14×14. Furthermore, $(\mathbf{X}^T\mathbf{X})^T = \mathbf{X}^T(\mathbf{X}^T)^T = \mathbf{X}^T\mathbf{X}$, proving it is always symmetric.

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13 [Intermediate] • Target: Eigenvalues, Eigenvectors and Variance Maximization

A centered feature matrix $\mathbf{X}$ has shape $N \times 18$. What is the size and symmetry property of the sample covariance matrix $\mathbf{\Sigma} = \frac{1}{N-1}\mathbf{X}^T\mathbf{X}$? Show your complete algebraic work.

Final Answer: 18×18 symmetric matrix. Since $\mathbf{X}$ has 18 columns, $\mathbf{X}^T$ is $18 \times N$ and $\mathbf{X}^T\mathbf{X}$ is 18×18. Furthermore, $(\mathbf{X}^T\mathbf{X})^T = \mathbf{X}^T(\mathbf{X}^T)^T = \mathbf{X}^T\mathbf{X}$, proving it is always symmetric.

14 [Intermediate] • Target: Explained Variance and Scree Analysis

A centered feature matrix $\mathbf{X}$ has shape $N \times 22$. What is the size and symmetry property of the sample covariance matrix $\mathbf{\Sigma} = \frac{1}{N-1}\mathbf{X}^T\mathbf{X}$? Show your complete algebraic work.

Final Answer: 22×22 symmetric matrix. Since $\mathbf{X}$ has 22 columns, $\mathbf{X}^T$ is $22 \times N$ and $\mathbf{X}^T\mathbf{X}$ is 22×22. Furthermore, $(\mathbf{X}^T\mathbf{X})^T = \mathbf{X}^T(\mathbf{X}^T)^T = \mathbf{X}^T\mathbf{X}$, proving it is always symmetric.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: SVD Connection and Applications • Cross-Lesson: 1.01 Vectors and Embeddings in AI

A centered feature matrix $\mathbf{X}$ has shape $N \times 26$. What is the size and symmetry property of the sample covariance matrix $\mathbf{\Sigma} = \frac{1}{N-1}\mathbf{X}^T\mathbf{X}$? Show your complete algebraic work.

Final Answer: 26×26 symmetric matrix. Since $\mathbf{X}$ has 26 columns, $\mathbf{X}^T$ is $26 \times N$ and $\mathbf{X}^T\mathbf{X}$ is 26×26. Furthermore, $(\mathbf{X}^T\mathbf{X})^T = \mathbf{X}^T(\mathbf{X}^T)^T = \mathbf{X}^T\mathbf{X}$, proving it is always symmetric.

16 [Challenging] • Target: PCA & Dimensionality Reduction Practice Exercises • Cross-Lesson: 2.01 Neural Layer Matrices and Activations

A centered feature matrix $\mathbf{X}$ has shape $N \times 30$. What is the size and symmetry property of the sample covariance matrix $\mathbf{\Sigma} = \frac{1}{N-1}\mathbf{X}^T\mathbf{X}$? Show your complete algebraic work.

Final Answer: 30×30 symmetric matrix. Since $\mathbf{X}$ has 30 columns, $\mathbf{X}^T$ is $30 \times N$ and $\mathbf{X}^T\mathbf{X}$ is 30×30. Furthermore, $(\mathbf{X}^T\mathbf{X})^T = \mathbf{X}^T(\mathbf{X}^T)^T = \mathbf{X}^T\mathbf{X}$, proving it is always symmetric.

17 [Challenging] • Target: Data Centering and the Covariance Matrix • Cross-Lesson: 3.01 Gradient Descent and Optimization

A centered feature matrix $\mathbf{X}$ has shape $N \times 34$. What is the size and symmetry property of the sample covariance matrix $\mathbf{\Sigma} = \frac{1}{N-1}\mathbf{X}^T\mathbf{X}$? Show your complete algebraic work.

Final Answer: 34×34 symmetric matrix. Since $\mathbf{X}$ has 34 columns, $\mathbf{X}^T$ is $34 \times N$ and $\mathbf{X}^T\mathbf{X}$ is 34×34. Furthermore, $(\mathbf{X}^T\mathbf{X})^T = \mathbf{X}^T(\mathbf{X}^T)^T = \mathbf{X}^T\mathbf{X}$, proving it is always symmetric.

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18 [Challenging] • Target: Eigenvalues, Eigenvectors and Variance Maximization • Cross-Lesson: 4.01 Conditional Probability and Bayes Rule

A centered feature matrix $\mathbf{X}$ has shape $N \times 38$. What is the size and symmetry property of the sample covariance matrix $\mathbf{\Sigma} = \frac{1}{N-1}\mathbf{X}^T\mathbf{X}$? Show your complete algebraic work.

Final Answer: 38×38 symmetric matrix. Since $\mathbf{X}$ has 38 columns, $\mathbf{X}^T$ is $38 \times N$ and $\mathbf{X}^T\mathbf{X}$ is 38×38. Furthermore, $(\mathbf{X}^T\mathbf{X})^T = \mathbf{X}^T(\mathbf{X}^T)^T = \mathbf{X}^T\mathbf{X}$, proving it is always symmetric.

19 [Challenging] • Target: Explained Variance and Scree Analysis • Cross-Lesson: 5.01 Distributions and Maximum Likelihood

A centered feature matrix $\mathbf{X}$ has shape $N \times 42$. What is the size and symmetry property of the sample covariance matrix $\mathbf{\Sigma} = \frac{1}{N-1}\mathbf{X}^T\mathbf{X}$? Show your complete algebraic work.

Final Answer: 42×42 symmetric matrix. Since $\mathbf{X}$ has 42 columns, $\mathbf{X}^T$ is $42 \times N$ and $\mathbf{X}^T\mathbf{X}$ is 42×42. Furthermore, $(\mathbf{X}^T\mathbf{X})^T = \mathbf{X}^T(\mathbf{X}^T)^T = \mathbf{X}^T\mathbf{X}$, proving it is always symmetric.

20 [Challenging] • Target: SVD Connection and Applications • Cross-Lesson: 6.01 Linear Regression and OLS

Consider a 2D data point $\mathbf{v} = [3;\ -4]$. What is the squared Euclidean norm (or squared magnitude) of this vector? Show your complete algebraic work.

Final Answer: 25. The squared Euclidean norm of a vector $\mathbf{v} = [v_1; v_2]$ is given by $||\mathbf{v}||^2 = v_1^2 + v_2^2$. For the given vector $\mathbf{v} = [3;\ -4]$, the squared Euclidean norm is $(3)^2 + (-4)^2 = 9 + 16 = 25$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises

Principal Component Analysis (PCA) projects a dataset from \(\mathbb{R}^{30} \) onto its top \(2 \) principal components. What is the dimension of each transformed data point?

a. \(15 \)

b. \(30 \)

c. \(28 \)

d. **\(2 \)** ✓

PCA projection onto \(k \) eigenvectors transforms each sample vector from \(\mathbb{R}^D \) to \(\mathbb{R}^k \). Here \(k = 2 \), so the reduced dimension is \(2 \).

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22 [Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises

Principal Component Analysis (PCA) projects a dataset from $\mathbb{R}^{40}$ onto its top 4 principal components. What is the dimension of each transformed data point?

a. 20 **b.** 40

c. 36 **d.** **4** ✓

PCA projection onto k eigenvectors transforms each sample vector from $\mathbb{R}^D$ to $\mathbb{R}^k$. Here $k = 4$, so the reduced dimension is 4.

23 [Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises

Principal Component Analysis (PCA) projects a dataset from $\mathbb{R}^{50}$ onto its top 6 principal components. What is the dimension of each transformed data point?

a. 25 **b.** 50

c. 44 **d.** **6** ✓

PCA projection onto k eigenvectors transforms each sample vector from $\mathbb{R}^D$ to $\mathbb{R}^k$. Here $k = 6$, so the reduced dimension is 6.

24 [Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises

Principal Component Analysis (PCA) projects a dataset from $\mathbb{R}^{60}$ onto its top 8 principal components. What is the dimension of each transformed data point?

a. 30 **b.** 60

c. 52 **d.** **8** ✓

PCA projection onto k eigenvectors transforms each sample vector from $\mathbb{R}^D$ to $\mathbb{R}^k$. Here $k = 8$, so the reduced dimension is 8.

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25 [Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises

Principal Component Analysis (PCA) projects a dataset from \(\mathbb{R}^{70} \) onto its top \(10 \) principal components. What is the dimension of each transformed data point?

a. **\(10 \) ✓**

b. \(70 \)

c. \(60 \)

d. \(35 \)

PCA projection onto \(k \) eigenvectors transforms each sample vector from \(\mathbb{R}^D \) to \(\mathbb{R}^k \). Here \(k = 10 \), so the reduced dimension is \(10 \).

26 [Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises

Principal Component Analysis (PCA) projects a dataset from \(\mathbb{R}^{80} \) onto its top \(12 \) principal components. What is the dimension of each transformed data point?

a. **\(12 \) ✓**

b. \(80 \)

c. \(68 \)

d. \(40 \)

PCA projection onto \(k \) eigenvectors transforms each sample vector from \(\mathbb{R}^D \) to \(\mathbb{R}^k \). Here \(k = 12 \), so the reduced dimension is \(12 \).

27 [Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises

Principal Component Analysis (PCA) projects a dataset from \(\mathbb{R}^{90} \) onto its top \(14 \) principal components. What is the dimension of each transformed data point?

a. **\(14 \) ✓**

b. \(90 \)

c. \(76 \)

d. \(45 \)

PCA projection onto \(k \) eigenvectors transforms each sample vector from \(\mathbb{R}^D \) to \(\mathbb{R}^k \). Here \(k = 14 \), so the reduced dimension is \(14 \).

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28 [Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises

Principal Component Analysis (PCA) projects a dataset from $\mathbb{R}^{100}$ onto its top 16 principal components. What is the dimension of each transformed data point?

a. 100

b. **16 ✓**

c. 84

d. 50

PCA projection onto k eigenvectors transforms each sample vector from $\mathbb{R}^D$ to $\mathbb{R}^k$. Here $k = 16$, so the reduced dimension is 16.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Data Centering and the Covariance Matrix

Why must data features be centered (subtracting feature means) prior to performing Principal Component Analysis?

a. **So the first principal component aligns with the axis of maximum variance rather than the offset from origin ✓**

b. To make the data non-linear

c. To ensure all eigenvalues are equal to 1

d. To reduce the matrix dimension

Without centering, the first component simply points toward the data mean rather than capturing internal variation.

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30 [Intermediate] • Target: Data Centering and the Covariance Matrix

For a zero-mean data matrix $\mathbf{X} \in \mathbf{R}^{N \times D}$, state the sample covariance matrix $\mathbf{\Sigma}$.

a. $\mathbf{\Sigma} = \frac{1}{N-1}\mathbf{X}^T\mathbf{X}$ ✓

b. $\mathbf{\Sigma} = \frac{1}{D-1}\mathbf{X}\mathbf{X}^T$

c. $\mathbf{\Sigma} = \mathbf{X}^T\mathbf{X}$

d. $\mathbf{\Sigma} = \frac{1}{N}\mathbf{X}^2$

Each element $\Sigma_{ij} = \frac{1}{N-1}\sum_k X_{ki}X_{kj}$, which in matrix form is $\frac{1}{N-1}\mathbf{X}^T\mathbf{X}$.

31 [Intermediate] • Target: Data Centering and the Covariance Matrix

What is the mathematical symmetry property of the covariance matrix $\mathbf{\Sigma}$?

a. **It is symmetric and positive semi-definite (all eigenvalues $\lambda_i \geq 0$)** ✓

b. It is skew-symmetric

c. It is an orthogonal matrix

d. It has negative trace

$\mathbf{\Sigma}^T = \left(\frac{1}{N-1}\mathbf{X}^T\mathbf{X}\right)^T = \frac{1}{N-1}\mathbf{X}^T\mathbf{X} = \mathbf{\Sigma}$, and variance is always non-negative.

32 [Intermediate] • Target: Eigenvalues, Eigenvectors and Variance Maximization

A 2×2 covariance matrix has entries $\Sigma_{11} = 4$, $\Sigma_{22} = 1$, and covariance $\Sigma_{12} = \Sigma_{21} = 2$. Find its eigenvalues.

a. $\lambda_1 = 5, \quad \lambda_2 = 0$ ✓

b. $\lambda_1 = 4, \quad \lambda_2 = 1$

c. $\lambda_1 = 3, \quad \lambda_2 = 2$

d. $\lambda_1 = 2, \quad \lambda_2 = 2$

$\det(\mathbf{\Sigma} - \lambda\mathbf{I}) = (4-\lambda)(1-\lambda) - 4 = \lambda^2 - 5\lambda = \lambda(\lambda - 5) = 0 \Rightarrow \lambda = 5, 0.$

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33 [Intermediate] • Target: Eigenvalues, Eigenvectors and Variance Maximization

What is the geometric meaning of the first eigenvector $\mathbf{v}_1$ corresponding to largest eigenvalue λ_1?

a. The direction along which the data exhibits maximum variance ✓

b. The direction perpendicular to all data points

c. The direction with the lowest information content

d. The vector of class centroids

PCA seeks $\max \mathbf{v}^T\mathbf{\Sigma}\mathbf{v}$ subject to $||\mathbf{v}|| = 1$. The solution is the eigenvector of $\lambda_{\max}$.

34 [Intermediate] • Target: Eigenvalues, Eigenvectors and Variance Maximization

If $\mathbf{v}$ is an eigenvector of $\mathbf{\Sigma}$ with eigenvalue λ, what is the variance of data projected onto $\mathbf{v}$?

a. λ ✓

b. $\sqrt{\lambda}$

c. λ^2

d. 1

$\text{Var}(\mathbf{X}\mathbf{v}) = \mathbf{v}^T\mathbf{\Sigma}\mathbf{v} = \mathbf{v}^T(\lambda\mathbf{v}) = \lambda||\mathbf{v}||^2 = \lambda$.

35 [Intermediate] • Target: Explained Variance and Scree Analysis

A dataset has eigenvalues $\lambda_1 = 70, \lambda_2 = 20, \lambda_3 = 10$. What percentage of total variance is explained by the first two principal components?

a. 90% ✓

b. 70%

c. 80%

d. 20%

Total variance $= 70 + 20 + 10 = 100$. Explained $= \frac{70+20}{100} = 90\%$.

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36 [Intermediate] • Target: Explained Variance and Scree Analysis

How is an 'elbow' in a scree plot (eigenvalues plotted against component index) used in practice?

a. To select the optimal cutoff k where adding further components yields diminishing returns ✓

b. To detect outliers in the target labels

c. To determine the learning rate for gradient descent

d. To test for multicollinearity

The elbow point separates significant signal dimensions from noise dimensions.

37 [Intermediate] • Target: Explained Variance and Scree Analysis

If a dataset has 100 features and you retain the top 10 principal components, by what percentage is feature dimensionality reduced?

a. 90% ✓

b. 10%

c. 50%

d. 80%

Reduction is from 100 to 10 features: $\frac{100-10}{100} = 90\%$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: SVD Connection and Applications • Cross-Lesson: 4.01 Conditional Probability and Bayes Rule

Given the Singular Value Decomposition $\mathbf{X} = \mathbf{USV}^T$, how are the principal component directions $\mathbf{v}_i$ directly obtained?

a. As the right singular vectors (columns of $\mathbf{V}$) ✓

b. As the left singular vectors (columns of $\mathbf{U}$)

c. As the diagonal elements of $\mathbf{S}$

d. As the eigenvalues of $\mathbf{U}$

$\mathbf{X}^T\mathbf{X} = \mathbf{VSU}^T\mathbf{USV}^T = \mathbf{VS}^2\mathbf{V}^T$, proving columns of $\mathbf{V}$ are eigenvectors of $\mathbf{X}^T\mathbf{X}$.

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39 [Challenging] • Target: SVD Connection and Applications • Cross-Lesson: 5.01 Distributions and Maximum Likelihood

What is the mathematical relation between singular values s_i of $\mathbf{X}$ and eigenvalues λ_i of covariance matrix $\mathbf{\Sigma}$?

a. $\lambda_i = \frac{s_i^2}{N-1}$ ✓

b. $\lambda_i = s_i$

c. $\lambda_i = \sqrt{s_i}$

d. $\lambda_i = s_i(N-1)$

Since $\mathbf{\Sigma} = \frac{1}{N-1}\mathbf{X}^T\mathbf{X}$ and eigenvalues of $\mathbf{X}^T\mathbf{X}$ are s_i^2, $\lambda_i = \frac{s_i^2}{N-1}$.

40 [Challenging] • Target: SVD Connection and Applications • Cross-Lesson: 6.01 Linear Regression and OLS

Name a major practical application of PCA in image processing and computer vision.

a. **Eigenfaces: facial compression and feature representation in lower-dimensional eigenspace** ✓

b. Image sharpening through edge convolution

c. Color palette inversion

d. Bilinear image interpolation

Eigenfaces (Turk and Pentland) represents high-resolution faces as linear combinations of dominant eigenvectors.

41 [Challenging] • Target: PCA & Dimensionality Reduction Practice Exercises • Cross-Lesson: 7.01 Logistic Regression and Classification

Principal Component Analysis (PCA) projects a dataset from $\mathbb{R}^{50}$ onto its top 6 principal components. What is the dimension of each transformed data point?

a. 25

b. 50

c. 44

d. 6 ✓

PCA projection onto k eigenvectors transforms each sample vector from $\mathbb{R}^D$ to $\mathbb{R}^k$. Here $k = 6$, so the reduced dimension is 6.

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42 [Challenging] • Target: Data Centering and the Covariance Matrix • Cross-Lesson: 1.01 Vectors and Embeddings in AI

Consider a linear transformation represented by the matrix $A = (3, 0; 0, -2)$. If $\mathbf{v} = (1; 1)$ is an eigenvector of A, what is the corresponding eigenvalue?

a. **3** ✓

b. -2

c. 1

d. 5

An eigenvector $\mathbf{v}$ of a matrix A satisfies the equation $A\mathbf{v} = \lambda\mathbf{v}$, where λ is the corresponding eigenvalue. We are given the matrix $A = (3, 0; 0, -2)$ and an eigenvector $\mathbf{v} = (1; 1)$. Let's compute $A\mathbf{v}$:

$A\mathbf{v} = (3, 0; 0, -2)(1; 1) = ((3)(1) + (0)(1); (0)(1) + (-2)(1)) = (3; -2)$ Now, we need to find λ such that $A\mathbf{v} = \lambda\mathbf{v}$, which means.

43 [Challenging] • Target: Eigenvalues, Eigenvectors and Variance Maximization • Cross-Lesson: 2.01 Neural Layer Matrices and Activations

Consider a dataset of n data points, where each data point is a vector in $\mathbb{R}^D$. A non-linear dimensionality reduction technique maps these data points to a lower-dimensional space $\mathbb{R}^d$. The mapping function for a single data point $\mathbf{x} \in \mathbb{R}^D$ is given by $f(\mathbf{x}) = \phi(\mathbf{x})$, where $\phi : \mathbb{R}^D \to \mathbb{R}^d$ is a function designed to preserve local neighborhood structures. Suppose the mapping ϕ is defined by an explicit feature map $\phi(\mathbf{x}) = [\mathbf{x}^T\mathbf{x}, \mathbf{x}^T\mathbf{x}, \ldots, \mathbf{x}^T\mathbf{x}]^T$ where the vector $\mathbf{x} \in \mathbb{R}^{10}$ and d is the dimension of the embedded space. If the goal is to embed these points into a space such that pairwise Euclidean distances are preserved as closely as possible under a specific distortion, and it is found that $d = 15$ yields optimal results for this mapping under a certain constraint, determine the effective rank of the feature map's image space.

a. 15

b. 10

c. **1** ✓

d. 25

The feature map is defined as $\phi(\mathbf{x}) = [\mathbf{x}^T\mathbf{x}, \mathbf{x}^T\mathbf{x}, \ldots, \mathbf{x}^T\mathbf{x}]^T$, where $\mathbf{x} \in \mathbb{R}^{10}$. The term $\mathbf{x}^T\mathbf{x}$ is the squared Euclidean norm of $\mathbf{x}$, denoted as $||\mathbf{x}||^2$. This is a scalar value. Thus, the mapping ϕ produces a vector where each component is the same scalar value, $||\mathbf{x}||^2$. Let $k = ||\mathbf{x}||^2$. Then $\phi(\mathbf{x}) = [k, k, \ldots, k]^T$ in $\mathbb{R}^d$. Th.

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44 [Challenging] • Target: Explained Variance and Scree Analysis • Cross-Lesson: 3.01 Gradient Descent and Optimization

The vector $\mathbf{v} = (-2; 1)$ is an eigenvector of the matrix $A = (3, 2; 1, 4)$. What is the corresponding eigenvalue λ?

a. 2 ✓

b. 5

c. -2

d. 7

By definition, a non-zero vector $\mathbf{v}$ is an eigenvector of A corresponding to eigenvalue λ if and only if $A\mathbf{v} = \lambda\mathbf{v}$. Compute the matrix-vector product $A\mathbf{v}$: $A\mathbf{v} = (3, 2; 1, 4)(-2; 1) = (3(-2) + 2(1); 1(-2) + 4(1)) = (-6 + 2; -2 + 4) = (-4; 2)$ Comparing $A\mathbf{v}$ to $\lambda\mathbf{v}$: $(-4; 2) = \lambda(-2; 1)$ This gives $-4 = -2\lambda$ and $2 = 1\lambda$, which.

45 [Challenging] • Target: SVD Connection and Applications • Cross-Lesson: 4.01 Conditional Probability and Bayes Rule

Consider a 2×2 matrix A with an eigenvalue $\lambda = 3$ and a corresponding eigenvector $\mathbf{v} = (1; 2)$. If the trace of A is 7, find the other eigenvalue of A.

a. 1

b. **4** ✓

c. 10

d. 21

Let the matrix A be $(a, b; c, d)$. The eigenvalues of A are the roots of the characteristic equation $\det(A - \lambda I) = 0$. We are given that $\lambda_1 = 3$ is an eigenvalue. The trace of a matrix is the sum of its diagonal elements, so $\text{tr}(A) = a + d$. We are given that $\text{tr}(A) = 7$. The trace of a matrix is also equal to the sum of its eigenvalues. Let the eigenvalues be λ_1 and λ_2. Therefore, $\lambda_1 + \lambda_2 = \text{tr}(A)$. Substituting the given.

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